

CL420 Introduction to Biochemical Engineering
Department of Chemical Eng., IIT Bombay

Quiz 1

17 Aug 2026

Instructions:

- You can bring a cheat sheet (two A4 sides) written in your own handwriting. Photocopy not permitted.
 - The cheat sheet must include your name and roll no. and must be submitted after the quiz
 - Mobile phones, smartwatches, and other electronic devices are strictly prohibited
 - All questions are compulsory.
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1. Yeast grows almost as fast and as easily as bacteria. So why do companies still use CHO or HEK cells to make complex drugs like monoclonal antibodies, instead of just using yeast? **[1 mark]**

Solution: Yeast and bacteria can't add the correct Post translational modifications like glycosylation or form disulfide bonds properly. Mammalian cells like CHO/HEK can do this, which is needed for the drug to work safely and effectively in the human body. Mabs require Post-translation modification - glycosylation, disulfide bonds which are present in CHO and HEK. Yeast and bacteria do not have such properties.

2. Why can a "more reduced" carbon source (like glycerol) give a higher cell yield than a "less reduced" one (like acetic acid), when growing the same organism aerobically? **[1 mark]**

Solution: A more reduced substrate has more electrons packed into it, which the cell can extract as extra energy (more NADH/ATP) during breakdown. This extra energy generally lets the cell build more biomass per gram of substrate consumed. glycerol –
More reduced substrate → more electrons → more energy → high $Y_{s/x}$

3. A cell has thousands of different mRNA molecules but only a few types of rRNA. Why this big difference? [1 mark]

Solution: rRNA is a structural building-block of the ribosome, and every ribosome uses the exact same rRNA, so only one type is needed — just in huge quantities. mRNA carries the instructions for each individual protein, so the cell needs a different mRNA for every different protein it wants to make.

4. A fed-batch mammalian culture shows a small increase in agitation rate that increases oxygen transfer but also changes the product glycosylation profile. Discuss this in the context of CPP–CQA framework. [1 mark]

Solution: Agitation is a CPP and glycosylation is a CQA.

CPP (Critical Process Parameter): A process parameter whose variation can affect product quality. Here, agitation rate affects oxygen transfer and subsequently influences glycosylation.

CQA (Critical Quality Attribute): A measurable property of the product that must remain within an appropriate range to ensure product quality. Glycosylation profile is a product quality attribute

Agitation is a CPP - varying this affects oxygen transfer and influences glycosylation and glycosylation is a CQA - measurable property of product which is to be in specific range for good product quality.

5. Why does oxygen limitation generally reduce the biomass yield on glucose in an aerobic culture? [1 mark]

Solution. Less ATP is generated per glucose and more carbon can be diverted to by-products. Under oxygen limitation, oxidative phosphorylation becomes restricted, so cells generate less ATP per molecule of glucose. They may rely more on glycolysis and fermentative pathways, producing by-products such as lactate or ethanol rather than efficiently converting glucose carbon into biomass.

Yes, oxygen limitation – Oxidative phosphorylation limits – less ATP per glucose. Then the fermentation pathway activates – glucose is converted to lactate or ethanol. Thus the yield reduces.

6. A batch culture consumes 15 g of glucose and produces 5.4 g of new cell mass. Given:
Biomass = $\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$. **[1 mark]**

(a) Calculate the observed growth yield coefficient $Y_{x/s}$ (g cells / g substrate)

$$Y_{x/s} = 5.4 \text{ g cells} / 15 \text{ g glucose} = 0.36 \text{ g cells} / \text{g glucose}$$

(b) Convert your answer to $Y_{x/s}$ in C-mol biomass / C-mol glucose

Using biomass molecular weight = 24.6 g / C-mol and glucose molecular weight = 30.0 g / C-mol:

$$\text{C-mol cells} = 5.4 / 24.6 = 0.2195 \text{ C-mol}$$

$$\text{C-mol glucose} = 15 / 30.0 = 0.5000 \text{ C-mol}$$

$$Y_{x/s} = 0.2195 / 0.5000 = 0.439 \text{ C-mol biomass} / \text{C-mol glucose}$$

7. Per C-mole of glucose consumed in aerobic growth, 0.45 C-mole of biomass ($\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$) is formed with no separate product **[2 marks]**

(a) Using a carbon balance alone, find the CO_2 yield $Y_{c/s}$ (C-mol/C-mol glucose).

Given $Y_{x/s} = 0.45$:

$$Y_{s,\text{CO}_2} = 1.0 - Y_{x/s} = 1.0 - 0.45 = 0.55 \text{ C-mol CO}_2 / \text{C-mol glucose}$$

(b) Using the degree-of-reduction balance, find the O_2 demand $Y_{o/s}$ (mol O_2 /C-mol glucose).

Given $\gamma_x = 4.20$, γ_s (glucose) = 4.0.

$$\text{Electron Balance: } \gamma_s = Y_{x/s}\gamma_x + 4 Y_{o/s}$$

$$4.0 = 0.45(4.20) + 4 Y_{o/s}$$

$$4.0 = 1.890 + 4 Y_{o/s}$$

$$Y_{o/s} = 2.110 / 4 = 0.528 \text{ mol O}_2 / \text{C-mol glucose}$$

8. A glycerol-limited aerobic chemostat ($D = 0.15 \text{ h}^{-1}$) is fed 20 g L^{-1} glycerol and 3 g L^{-1} NH_3 . Steady-state effluent: glycerol 0.20 g L^{-1} , NH_3 2.1 g L^{-1} , biomass 6.0 g L^{-1} (biomass = $\text{CH}_{1.8}\text{O}_{0.5}\text{N}_{0.2}$; glycerol $\text{C}_3\text{H}_8\text{O}_3$). **[2 marks]**

(a) Calculate volumetric rate of substrate uptake and biomass formation (q_s and q_x , respectively) in C-mol. $\text{L}^{-1} \cdot \text{h}^{-1}$

$$q_s = D(S_f - S)/FW_{gly} = 0.15(20 - 0.20)/30.7 = 0.0967 \text{ C-mol L}^{-1} \text{ h}^{-1}$$

$$q_x = D \times X / FW_x = 0.15 \times 6.0 / 24.6 = 0.0366 \text{ C-mol L}^{-1} \text{ h}^{-1}$$

(b) Calculate $Y_{x/s}$ in C-mol biomass / C-mol glycerol.

$$Y_{sx} = q_x / q_s = 0.378 \text{ C-mol biomass / C-mol glycerol}$$

*******Paper Ends*******